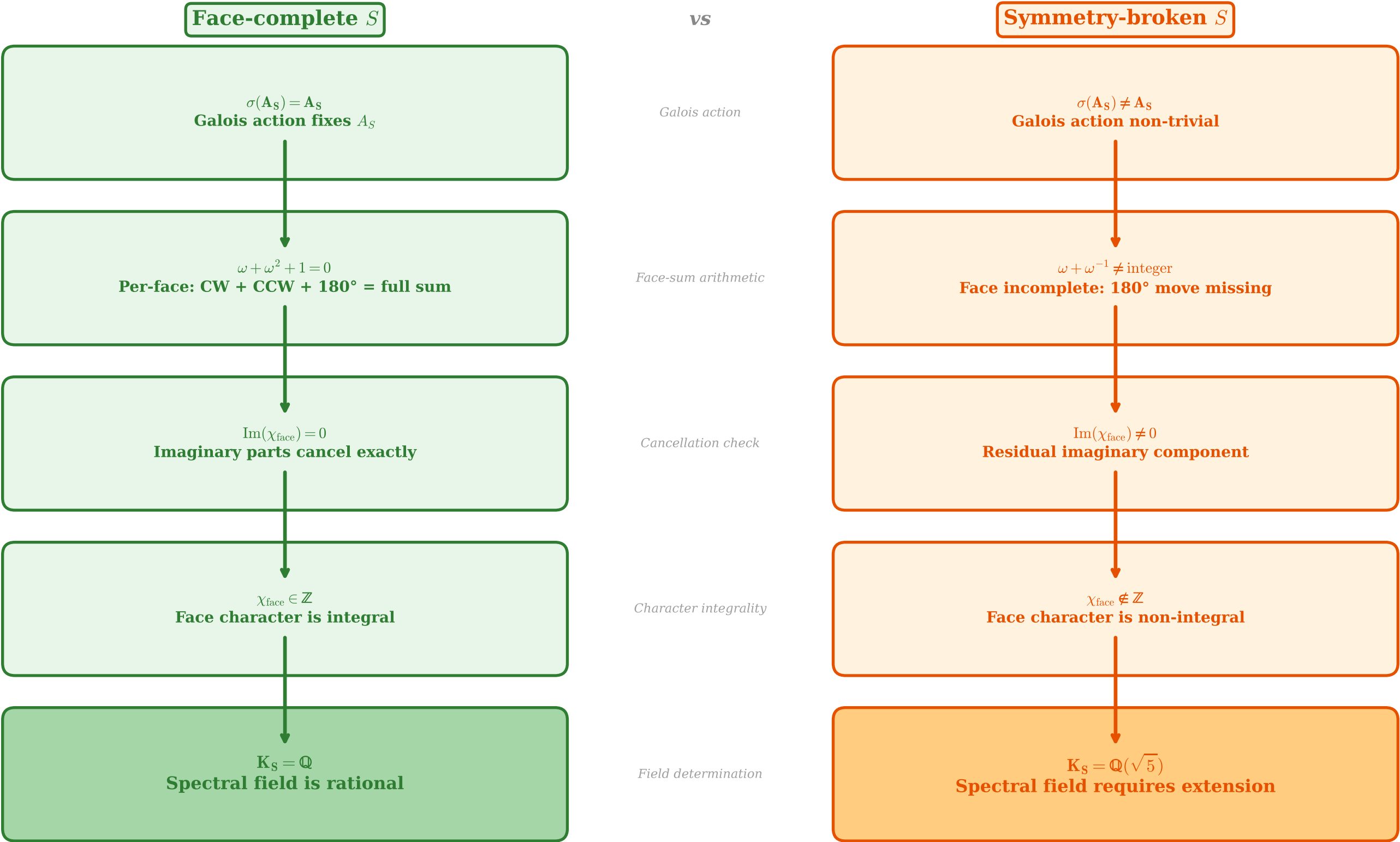


Galois Symmetry as the Mechanism of Spectral Rationality

Face-sum arithmetic determines the spectral field K_S



$\sigma(A) = A \Rightarrow K_S = \mathbb{Q}$ vs $\sigma(A) \neq A \Rightarrow K_S \supseteq \mathbb{Q}(\sqrt{5})$

Theorem (informal): Face-symmetry of S is necessary and sufficient for $K_S = \mathbb{Q}$